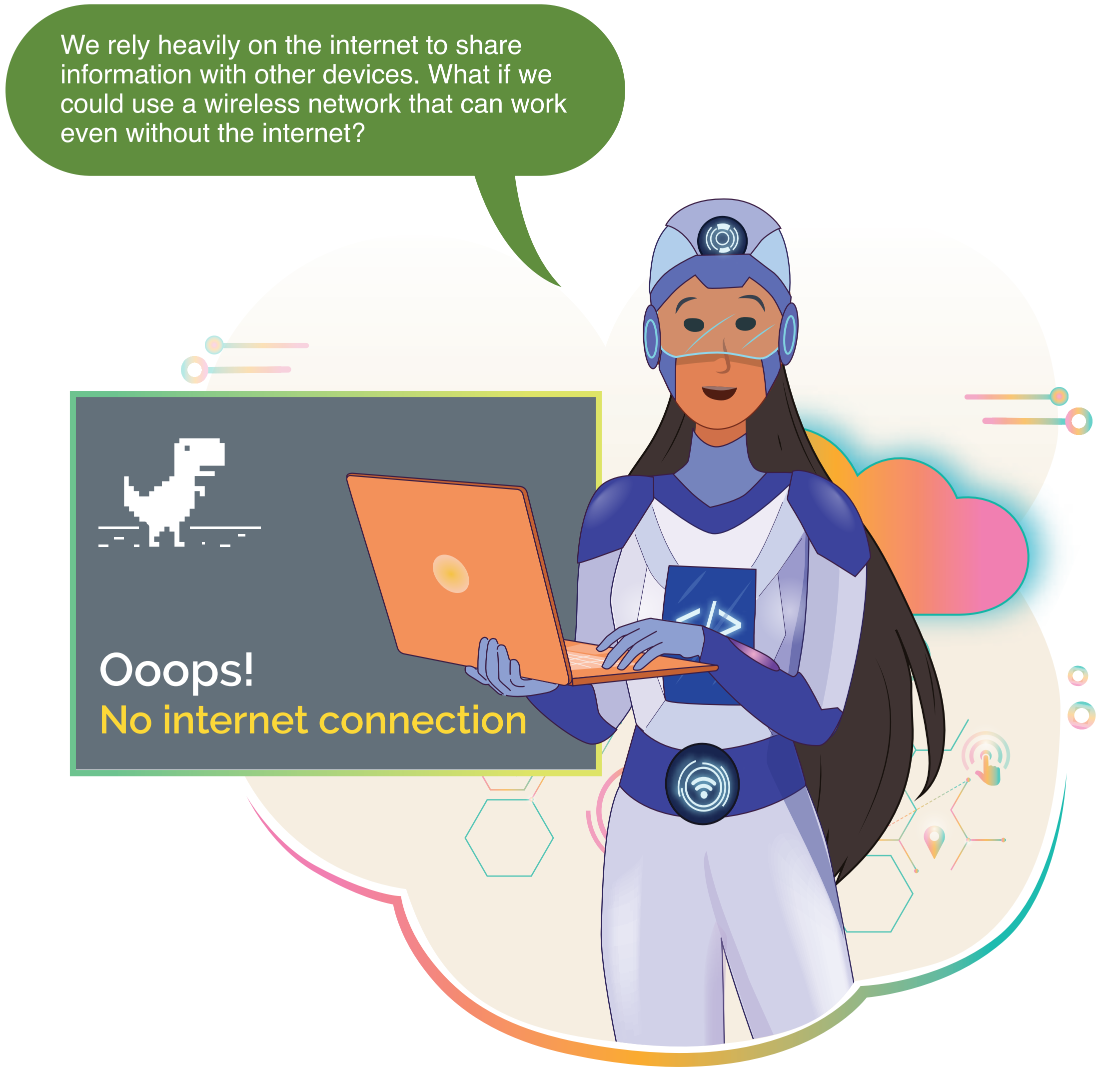
Bluetooth Communication 14



Over the years, computers and mobile phones have changed from simple communication devices to multi-functional tools. We live in a world where we seek products to ease our daily tasks. Applications make this possible! Which are some of your favourite apps? What do you use them for?

14. Bluetooth Communication

Application

An application, also known as an app, is an interactive software to perform specific tasks. Tasks include sending and receiving emails, marking a calendar, making calls, finding locations, and playing games. Applications work by interacting with the users. Based on user inputs, applications give an appropriate output.

Until a few years ago, we had to go outside to book a train ticket, find a taxi, shop or transfer money. This has changed now with the exponential growth of applications. Apps make these tasks easier and more convenient to complete. We can use a smartphone or laptop to do this from any location. Presently, apps are used in the areas of communication, education, social media, shopping, business, banking and more.

Data transfer is one of the most essential tasks that a smartphone performs.

a. Data can be in the form of documents, image files, audio files or even a text message. When we upgrade to a new smartphone, we transfer important offline data from the old phone to the new one.

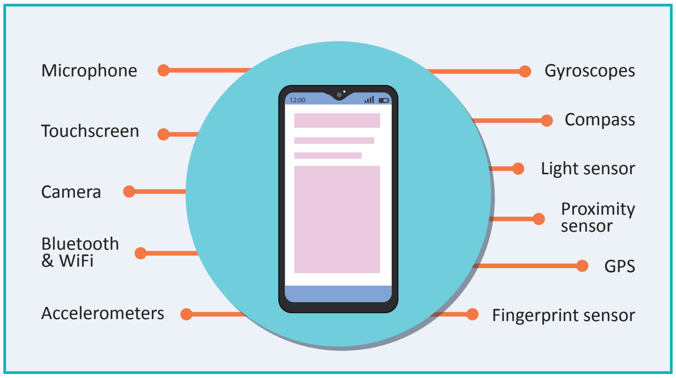
b. Different apps are available to transfer data from one phone to another, but they may incur extra charges.

c. One of the simplest methods to transfer any data from one device to another is through a Bluetooth sensor.

14. Bluetooth Communication

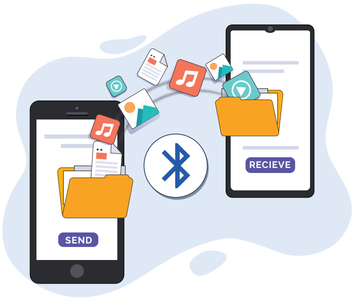
Sensors in a smartphone

Like other digital devices, smartphones have dozens of sensors built as per user requirements. Bluetooth is one such sensor that can be used for short distance communication.



Bluetooth

Bluetooth is an open wireless technology used to transfer data over short distances from a fixed or mobile device. It creates a highly secured Personal Area Network (PAN).

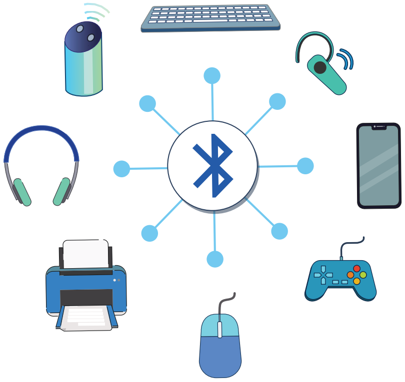


Bluetooth applications

a. Bluetooth slices data and transmits those pieces in a specific frequency range (2.4 GHz to 2.4835 GHz).

b. The pieces of information that go from one device to another are called "Packets".

c. Bluetooth enables a secure exchange of information between devices like faxes, smartphones, laptops, printers, speakers, digital cameras, video game consoles, etc.



Bytes

Bluetooth got its name from a Scandinavian king named Harald Gormsson. He was nicknamed 'Bluetooth' owing to a dead, dark blue tooth that he had. It was named after him because the King was famous for uniting Scandinavia, like Bluetooth technology unites devices.

14. Bluetooth Communication

Healthy Practices

a. Switch off a device's Bluetooth when not in use.

b. Do not pair your device with unknown devices.

How does a Bluetooth communication system work?

Bluetooth devices automatically detect and connect with each other. Up to eight of them can communicate at any one time. They don't interfere with one another because each pair of devices uses a different channel out of the 79 channels available.

If two devices want to communicate, they pick a random channel. If that channel is already occupied, they automatically switch to another.

When two or more devices communicate, one will act as a Bluetooth Server and others will act as a Bluetooth Client. The Bluetooth server transmits data to any of the connected clients. The clients can only transmit data to the server, not to each other.

Bluetooth devices should be "paired" first and establish a connection. The Bluetooth client sends a connection request, and the Bluetooth server accepts this request. Once they are paired, data can be transferred.

We can use an app development platform to create our own Bluetooth based communication app.

MIT App Inventor is a software used to create applications.

14. Bluetooth Communication

What is MIT App Inventor?

a. MIT App Inventor is an intuitive, visual programming environment that allows anyone to build fully functional apps for smartphones and tablets.

b. It uses block-based tools to create apps.

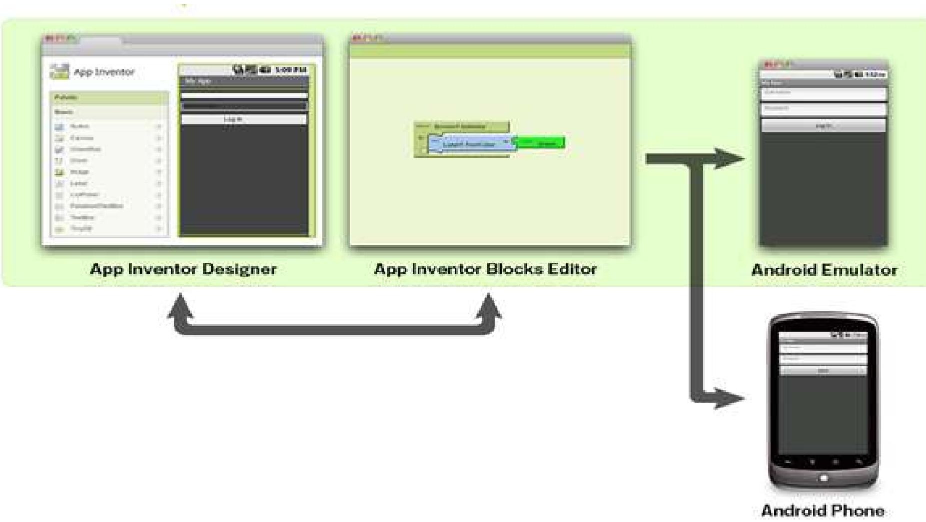
c. This method helps create complex and engaging apps in less time than traditional programming environments.

d. MIT app inventor can be accessed online via its website or offline via software. In both the cases, users login using a Gmail ID.

e. To open to the MIT app inventor online platform, open a browser window and enter the following link (internet required): https://appinventor.mit.edu/

The App Inventor consists of 3 components (windows) for programming:

1. Designer
2. Blocks Editor
3. Android Emulator or Android Phone

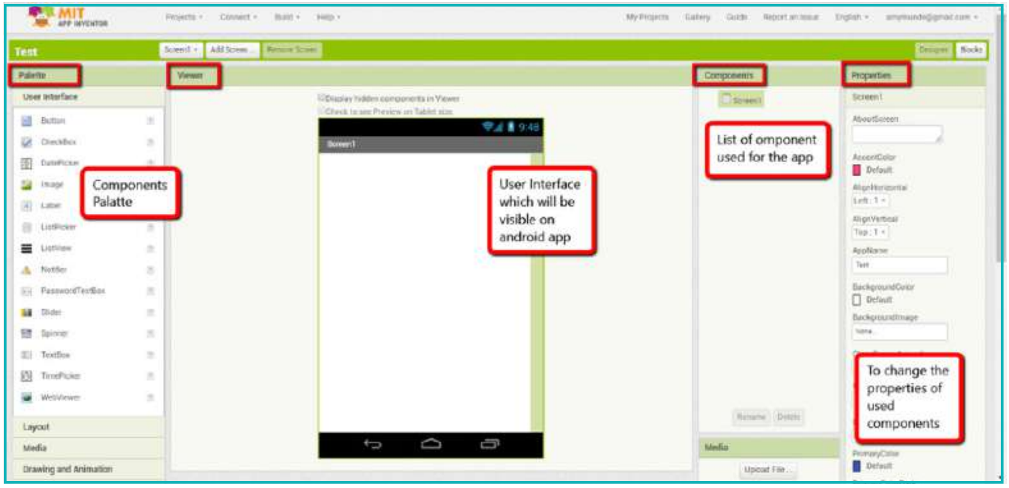


The double headed arrow in the image indicates that we can switch back and forth between two sections while designing an app.

To build an app, we need to use the app inventor designer and select application components. We also use this to design the User Interface (UI).

Understand the five sections of the design view shown in the image (Palette, Viewer, Components, Media and Properties).

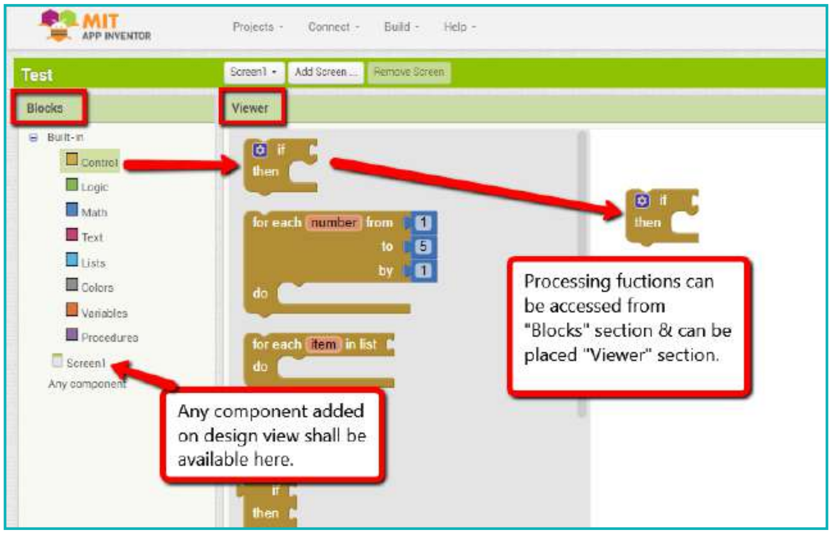
The palette and MIT app inventor both have different media sections.



a. To start designing a project, place a component on the designer screen.

b. To assemble program blocks, we use the app inventor blocks editor. Program blocks specify the functionality of components in the app.(internet required): https://appinventor.mit.edu/

c. There are two sections to the Blocks Editor: Blocks and Viewer.



d. This table explains the terms highlighted in the picture:

Sr.No.	Sub Parts (On Project Window)	Use
1	Designer	It is called User Interface (UI). We can create a visual representation of an app on the designer screen.
2	Blocks	Blocks are used to design the functionality of components that we place on the designer screen.
3	Palette	The palette holds all the components that we use to build an app.
4	Viewer	A component appears in the viewer as a graphic. We add components here to add them to the app.
5	Component	The component placed which we used in project, can be seen in “Component” section as a list of components.
6	Properties	We can set component settings in properties.

Exercise

State whether True or False

1. Bluetooth devices should be “paired” first and establish a connection.

   ____

2. If two devices want to communicate, they pick a random channel. If that channel is already occupied, no communication can happen.

   ____

3. When two or more devices communicate, one will act as a Bluetooth Server and others will act as Bluetooth Clients.

   ____

Activity

Create a Bluetooth communication system between two android smartphones.

To create a Bluetooth communication system between two android smartphones, we need to create 2 separate apps. One will act as a Bluetooth Server and the other as a Bluetooth Client. Each app will operate on a different smartphone.

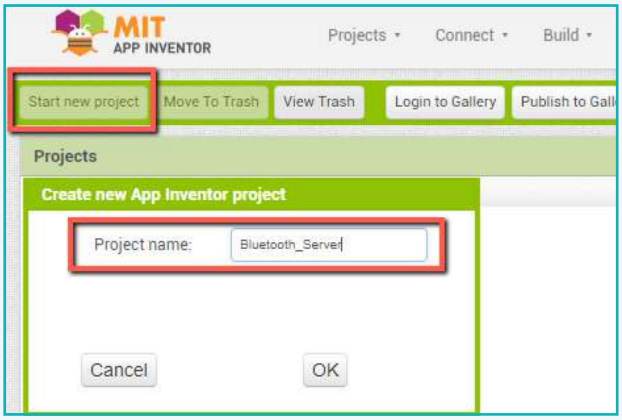
Implementing a screen for the Bluetooth server:

1. Go to web page: http://appinventor.mit.edu/explore/ai2/setup.html#

a. Click on create apps to start creating an application.

b. Click "continue" to dismiss the splash screen.

2. To start the project, click on "Start New Project". Name the project "Bluetooth_ Server" and click on "OK"



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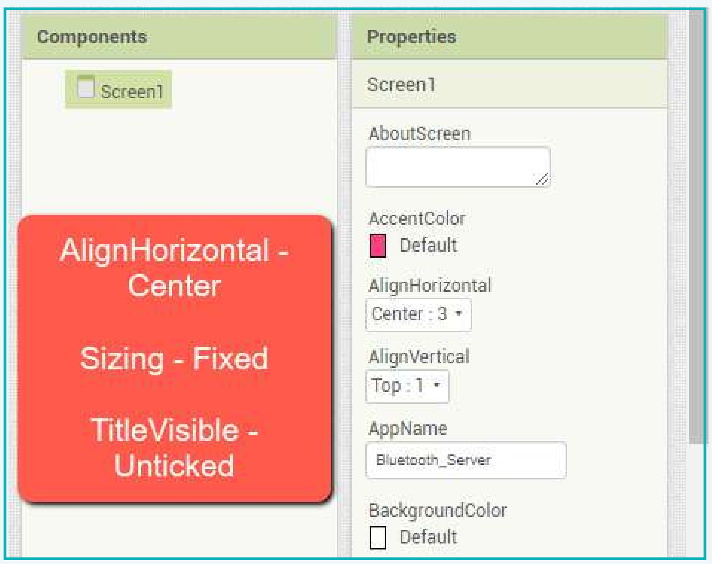
Note:

Do not use any spaces while naming the project.

3. Set screen properties as shown below:

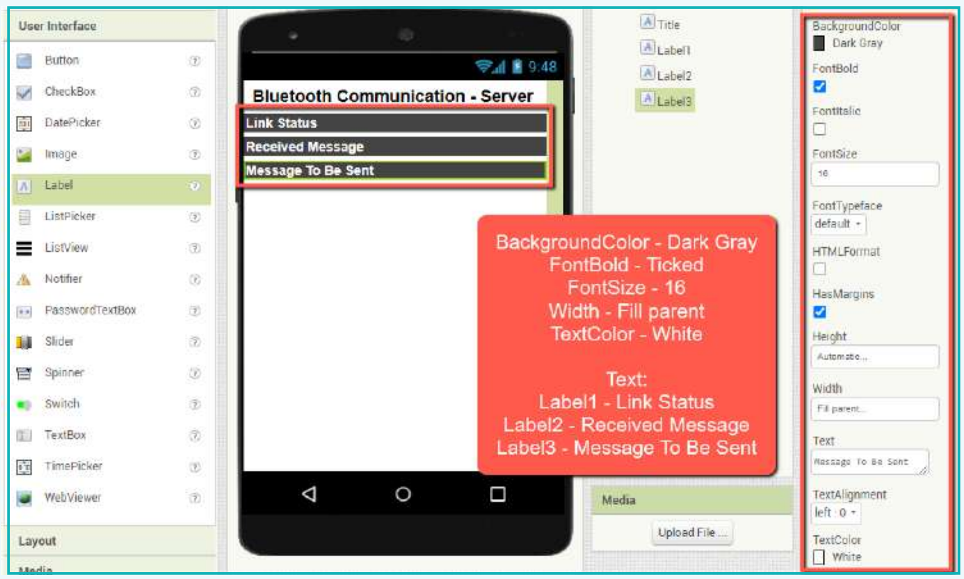


4. Add the following components to the Viewer section from the Palette. These will make up the user interface of the Bluetooth server app:

a. One label for the title

Activity

b. Three labels as “Status”, “Received Message” & “Message To Be Sent”. Set properties as shown in the image.



c. Add text box to display messages.

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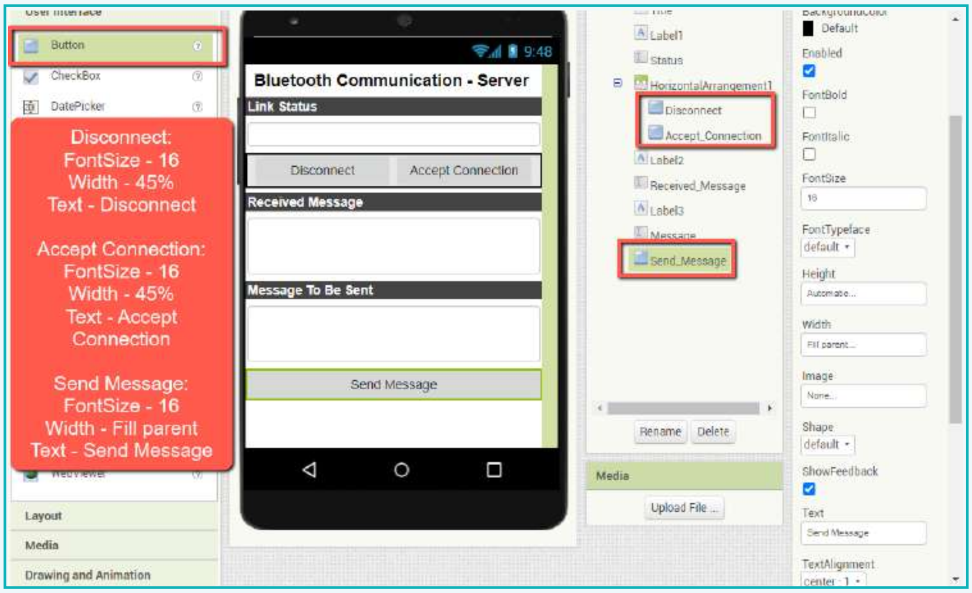
d. Add buttons for user inputs “Accept connection”, “Disconnect” & “Send Message”.



e. The final user interface for the Bluetooth Server app will look like this:

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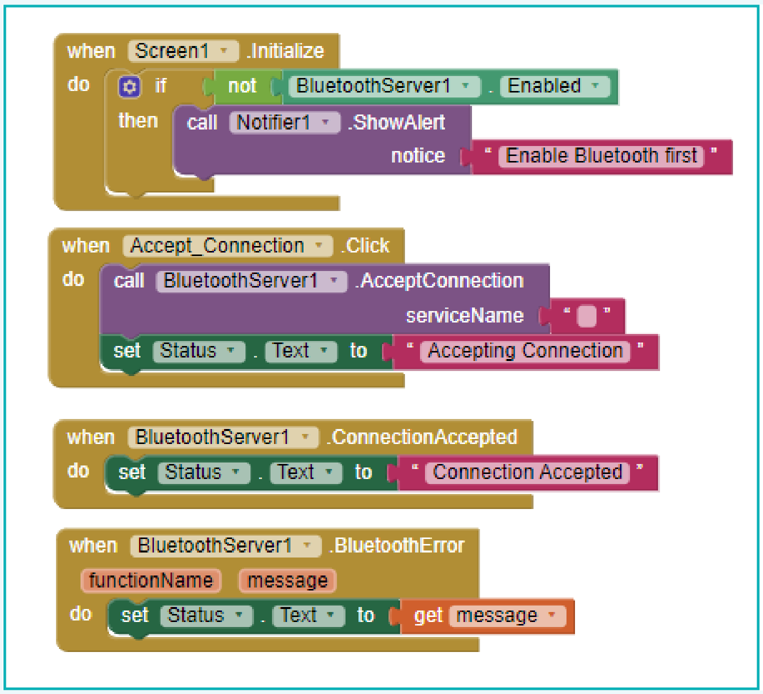
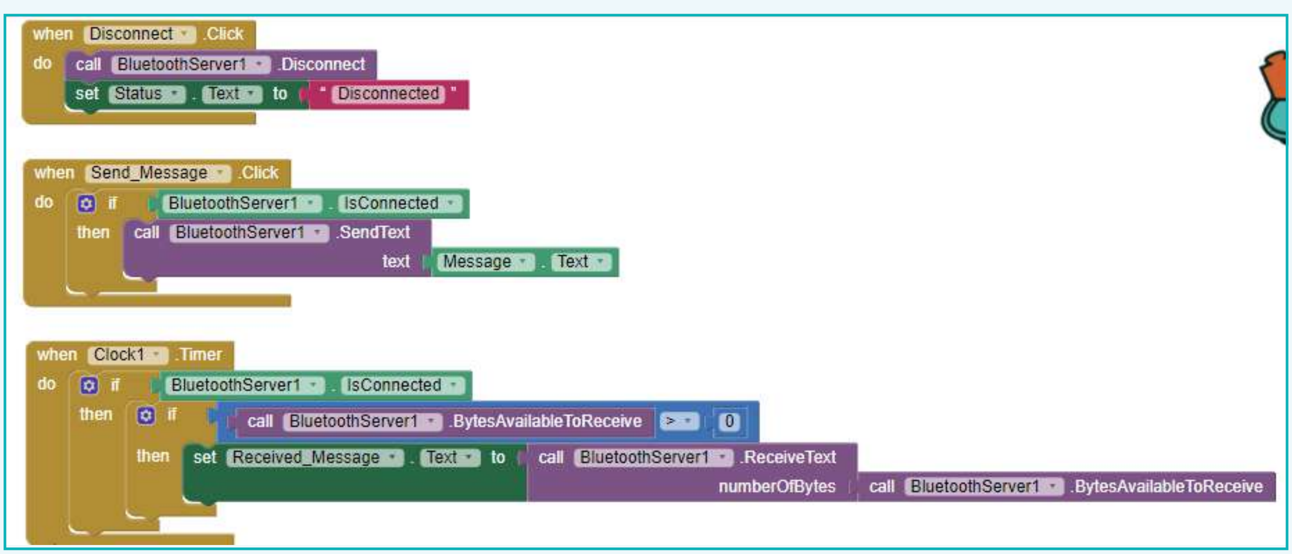
This is what the complete code for the Bluetooth Server application will look like:





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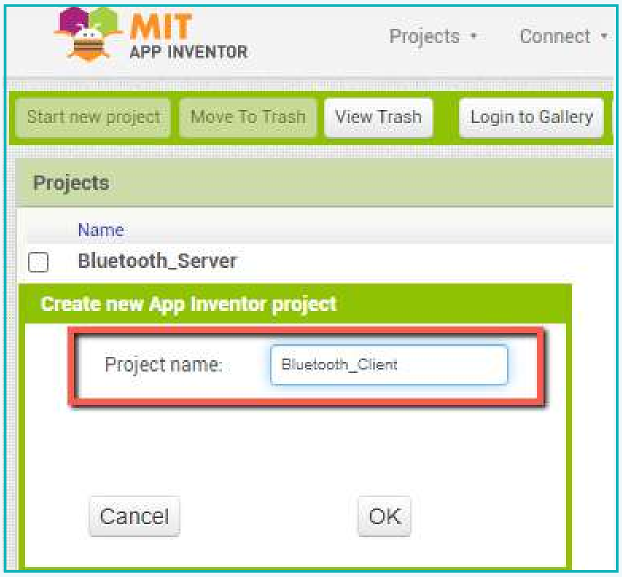
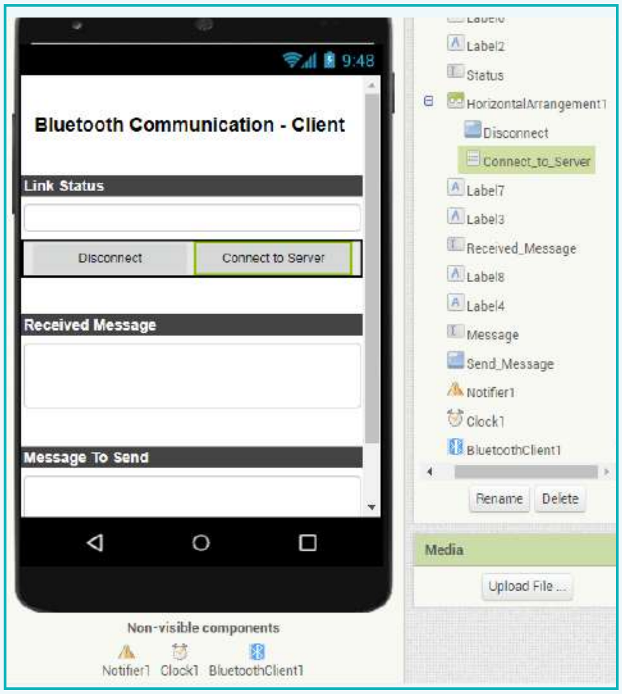
Implementing a screen for the Bluetooth client:

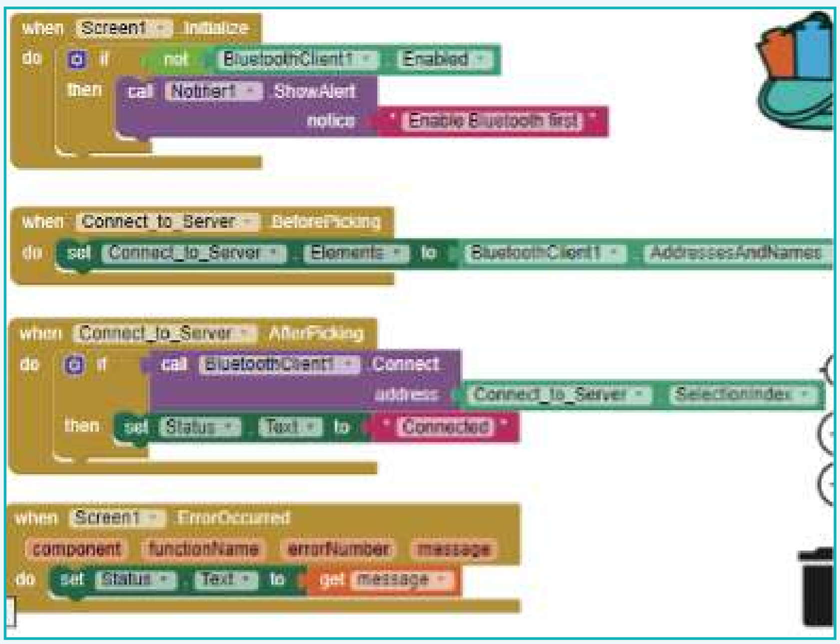
- Create a new app for the Bluetooth Client.

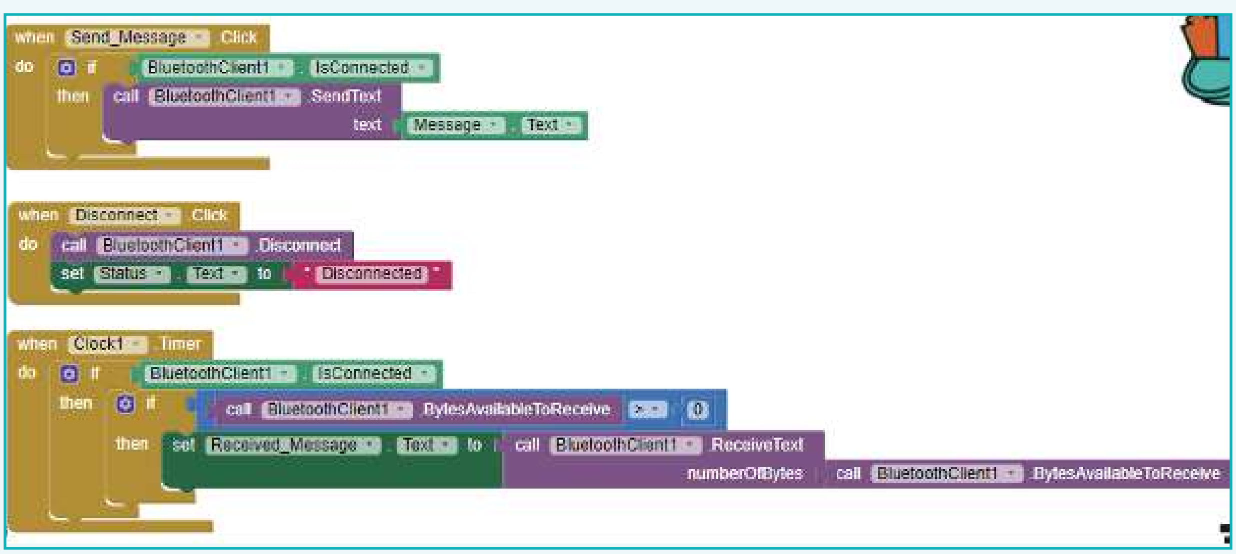




Activity

Code for Bluetooth Client:



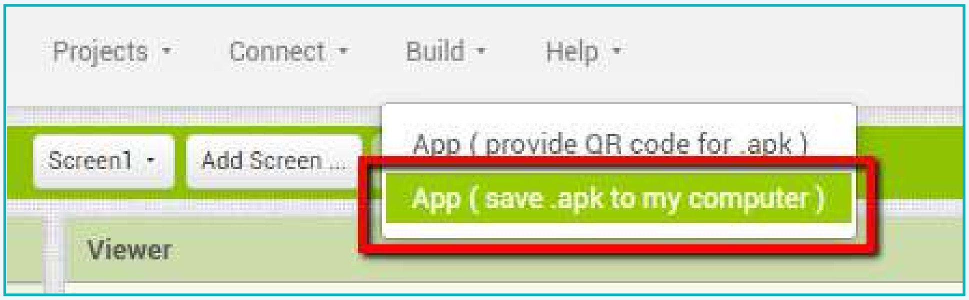


Activity

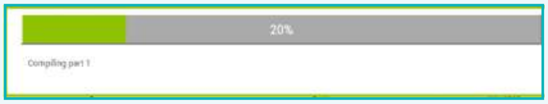
5 Test the app:

- **Emulator:** We use an android emulator to test the app on a PC, without needing any other device. This is useful when we don't have access to a smartphone while developing an app.
- **APK file:** We download the app's APK file on the computer. It is copied and installed on an android phone via a USB cable.

a. Go back to MIT App Inventor and click on "Build" at the top of the designer screen. Inside build, click on "App (save .apk to my computer)".



b. Wait for the process to be 100% complete.



c. Build APK files for both apps, Bluetooth Server and Bluetooth Client.

d. Copy the downloaded applications and paste them in separate Android phones.

e. Once we open the file on the phone, we see an option to install the app. Click on "Install" on both phones to install the applications.

f. Open both the applications together.

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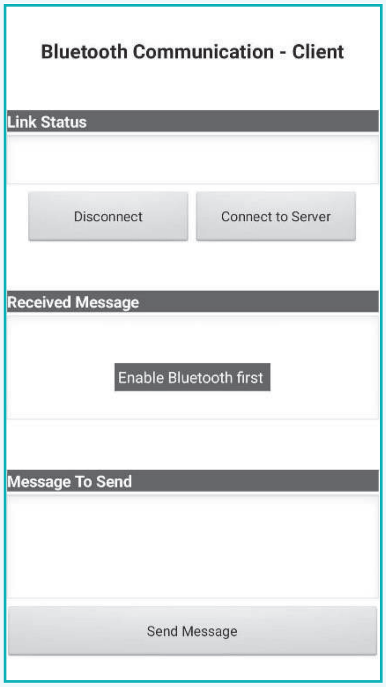
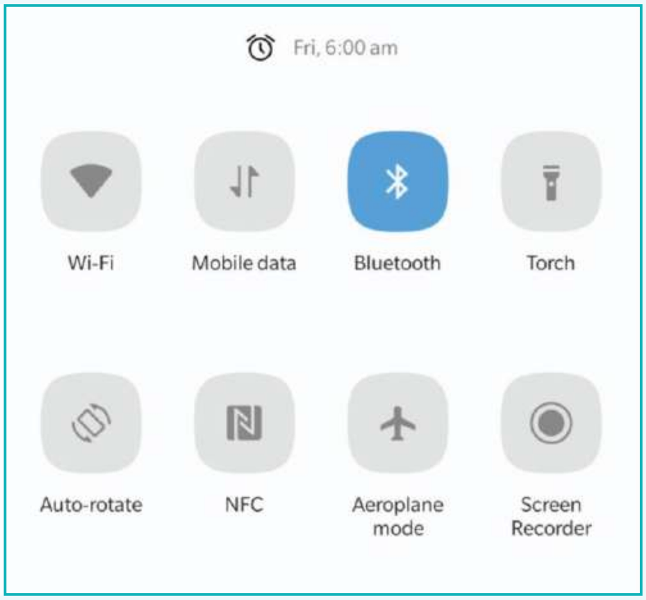
g. Both apps will display a notification saying, “Enable Bluetooth First”.



h. Enable Bluetooth on both devices.



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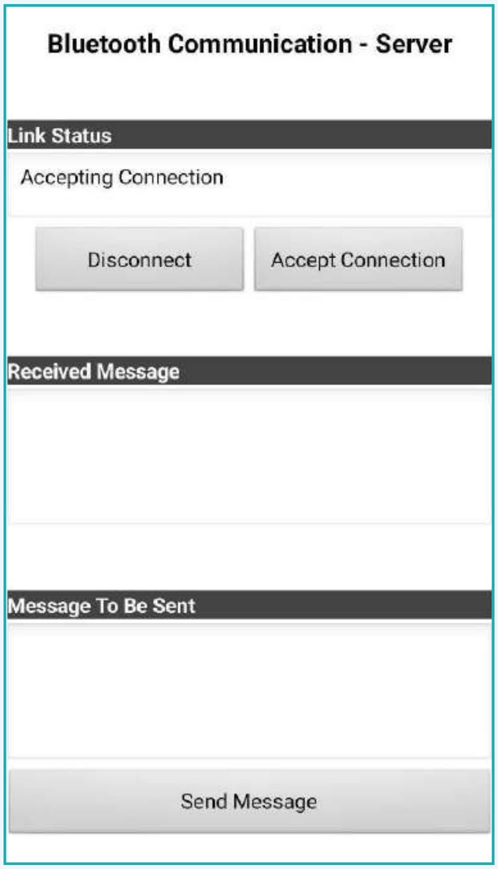
i. Click on 'Connect to the Server' to send a connection request from the client app to the server.

j. On the server app, click 'accept connection to pair' and connect the two devices.



k. Once the connection is established, both android phones can communicate with each other.

In this lesson, we learnt about the types of applications and how they make our lives comfortable. Every computer and mobile device come with a few built-in applications. Some applications can be downloaded inexpensively, or even for free. We also learnt all the steps involved in creating an app. What kind of app would you like to build? How would you want it to benefit people and create value?

14. Bluetooth Communication

Tech Flash

- **Bluetooth:** : Bluetooth is an open wireless technology used to transfer data over short distances from fixed or mobile devices.
- **Packets:** : Pieces of information that go from one device to another.
- **MIT App Inventor** : An intuitive, visual programming environment that allows anyone to build fully functional apps for smartphones and tablets.